SAHIL KUMAR

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ACADEMIC DETAILS

B.Tech (Mechanical Engineering)	2023-2027	MMMUT, Gorakhpur	6.64/10.0 (Ongoing)
Minor Degree (Data Science)	2024-2027	MMMUT, Gorakhpur	(Ongoing)
Class XII (BSEB)	2023	J.R.S Senior Secondary School, Siwan	75.4%
Class X (CBSE)	2021	Vatayan School, Siwan	61.4%

INTERNSHIP/TRAINING

Summer Training—Mechanical Workshop, N.E. Railway, Gorakhpur Link: *May 2025-June 2025*

- Studied the construction and working of LHB wheel and bearing system.
- Learned about wheel set assembly and bearing arrangement used in LHB coaches.
- Observed inspection, maintenance and safety practices followed in the workshop.
- Gained basic understanding of railway wheel and bearing components.

PROJECTS

Electrostatic Fog Water Collection System (Final Year Project). *Ongoing*

- Developing an electrostatic-assisted fog harvesting system to enhance water collection through controlled droplet dynamics and optimized surface design.
- Analyzing micro-droplet motion using principles of fluid mechanics and particle force balance.
- Investigating the influence of electrostatic fields on fog particle attraction and capture efficiency.
- Designing a conceptual prototype layout for experimental validation.
- Studying surface wettability and directional liquid transport mechanisms to improve droplet removal rate.

Dynamic Load Analysis Dashboard in Excel. *February-2025*

- **Project** Link**:**
- Analyzed ₹54.9+ Lakhs sales data and created an interactive Excel dashboard.
- Improved customer experience by identifying potential customers across different states, occupation, gender and age groups.
- Boosted sales by analyzing top-selling products to improve inventory management.

ACHIEVEMENTS

- **2nd Position – ELTIRO Technical Event, Techsrijan'24** Link**:** Designed and presented a functional air-combustion model using burning gases, demonstrating strong conceptual understanding of mechanical energy conversion and applied thermodynamics.
- Ranked **37th** in the **Internal Smart India Hackathon (SIH) 2025** Link **:** As part of Team Coffee Overflow, developing an AI-driven Tourist Safety & Incident Response System using Geo-fencing and Blockchain Digital ID.

SKILLS

Technical: SolidWorks, ANSYS, AutoCAD, MS PowerPoint, MS Word, MS Excel, MATLAB, Python (NumPy, Pandas, Matplotlib).

Curriculum: Heat and Mass Transfer, Refrigeration and Air Conditioning, Internal Combustion Engines, Engineering Graphics, Manufacturing Science & Manufacturing Processes, Mechanics of Solids, Numerical Methods.

Interests: Automobile Sector and Mechanical Design.
Soft Skills: Teamwork, Problem-Solving, Time Management, Adaptability.
Additional Skills: Data Analytics, Power BI, PostgreSQL, C++, Data Structures & Algorithms (DSA), HTML, CSS.
Certificates Link**:**

POSITION OF RESPONSIBILITY

- Core Team Member, ELTIRO Technical Event, Techsrijan'24 *(February 2024)* — Contributed to event planning, team coordination, and smooth execution of technical activities. *February-2024*
- Team Member, Smart India Hackathon (SIH)'25. September-2025

EXTRACURRICULAR

- Social Internship / Volunteering: Participated in community service initiatives, improving communication, leadership, and teamwork skills.
- Sports & Fitness: Regularly play Cricket and Badminton; actively maintain fitness through gym workouts, demonstrating discipline, perseverance, and teamwork.